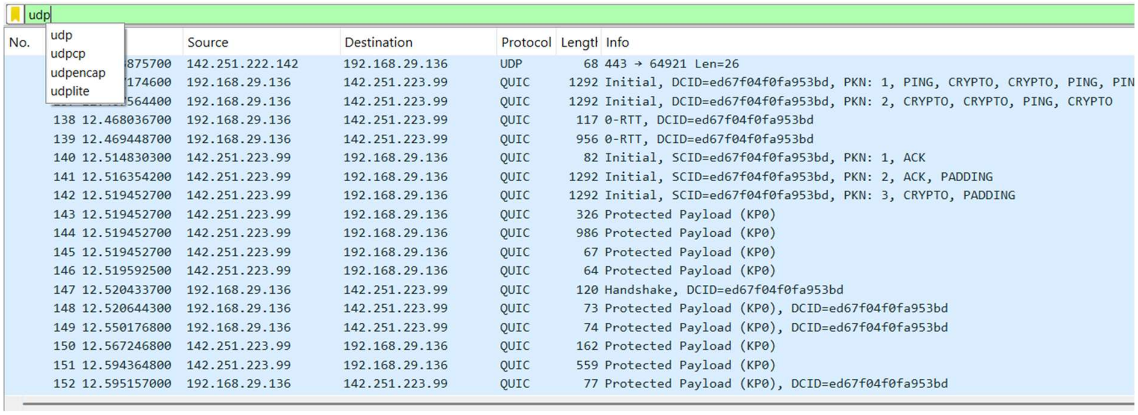
1. Try Test-Connection and nslookup commands for below websites
   www.google.com
   www.facebook.com
   www.amazon.com
   www.github.com
   www.cisco.com
2. Use Wireshark to capture and analyze DNS, TCP, UDP traffic and packet header, packet flow, options and flags
3. Explore traceroute/tracert for different websites eg:google.com and analyse the parameters in the output and explore different options for traceroute command
4. Use Cisco packet tracer for the below
5. Set up trunk ports between switches and try ping between different VLANs.
6. Change the native VLAN on a trunk port.Test for VLAN mismatches and troubleshoot.
7. Configure a management VLAN and assign an IP address for remote access.Test SSH or Telnet access to the switch.
8. You have a Cisco switch and a VoIP phone that needs to be placed in a voice VLAN (VLAN 20). The data for the PC should remain in a separate VLAN (VLAN 10). Configure the switch port to support both voice and data traffic.
9. You configured VLANs 10 and 20 on your switch and assigned ports to each VLAN. However, devices in VLAN 10 cannot communicate with devices in VLAN 20. Troubleshoot the issue.
10. Try Inter VLAN routing with Router
11. Implement ACLs to restrict traffic based on source and destination ports.Test rules by simulating legitimate and unauthorized traffic.
12. Configure a standard Access Control List (ACL) on a router to permit traffic from a specific IP range.Test connectivity to verify the ACL is working as intended.
13. Create an extended ACL to block specific applications, such as HTTP or FTP traffic.Test the ACL rules by attempting to access blocked services.
14. Try Static NAT, Dynamic NAT and PAT to translate IPs
15. Download iperf in laptop/phone and make sure they are in same network. Try different iperf commands with tcp, udp, birectional, reverse, multicast, parallel options and analyze the bandwidth and rate of transmission, delay, jitter etc.

1.

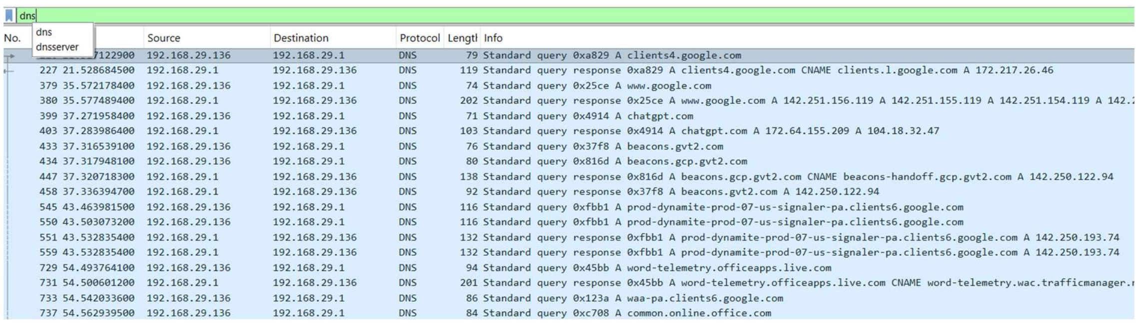
```
dharanish@DHARANISH-LOQ:~$ nslookup www.google.com
nslookup www.facebook.com
nslookup www.amazon.com
nslookup www.github.com
nslookup www.cisco.com
Server:         10.255.255.254
Address:        10.255.255.254#53

Non-authoritative answer:
Name:   www.google.com
Address: 142.251.156.119
Name:   www.google.com
Address: 142.251.157.119
Name:   www.google.com
Address: 142.251.154.119
Name:   www.google.com
Address: 142.251.150.119
Name:   www.google.com
Address: 142.251.153.119
Name:   www.google.com
Address: 142.251.155.119
Name:   www.google.com
Address: 142.251.152.119
Name:   www.google.com
Address: 142.251.151.119
Name:   www.google.com
Address: 2001:4860:482b:7700::
Name:   www.google.com
Address: 2001:4860:482d:7700::
```

```
dharanish@DHARANISH-LOQ:~$ ping -c 4 www.google.com
ping -c 4 www.facebook.com
ping -c 4 www.amazon.com
ping -c 4 www.github.com
ping -c 4 www.cisco.com
PING www.google.com (142.251.150.119) 56(84) bytes of data.
64 bytes from 142.251.150.119 (142.251.150.119): icmp_seq=1 ttl=111 time=20.2 ms
64 bytes from 142.251.150.119 (142.251.150.119): icmp_seq=2 ttl=111 time=15.6 ms
64 bytes from 142.251.150.119 (142.251.150.119): icmp_seq=3 ttl=111 time=15.7 ms
64 bytes from 142.251.150.119 (142.251.150.119): icmp_seq=4 ttl=111 time=15.5 ms

--- www.google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3188ms
rtt min/avg/max/mdev = 15.469/16.751/20.152/1.966 ms
PING star-mini.c10r.facebook.com (57.144.48.1) 56(84) bytes of data.
64 bytes from edge-star-mini-shv-02-del3.facebook.com (57.144.48.1): icmp_seq=1 ttl=49 time=49.3 ms
64 bytes from edge-star-mini-shv-02-del3.facebook.com (57.144.48.1): icmp_seq=2 ttl=49 time=53.8 ms
64 bytes from edge-star-mini-shv-02-del3.facebook.com (57.144.48.1): icmp_seq=3 ttl=49 time=48.7 ms
64 bytes from edge-star-mini-shv-02-del3.facebook.com (57.144.48.1): icmp_seq=4 ttl=49 time=55.4 ms

--- star-mini.c10r.facebook.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3179ms
rtt min/avg/max/mdev = 48.706/51.802/55.419/2.864 ms
PING cf.47cf2c8c9-frontier.amazon.com (18.67.150.142) 56(84) bytes of data.
64 bytes from server-18-67-150-142.maa51.r.cloudfront.net (18.67.150.142): icmp_seq=1 ttl=242 time=13.3 ms
64 bytes from server-18-67-150-142.maa51.r.cloudfront.net (18.67.150.142): icmp_seq=2 ttl=242 time=14.9 ms
64 bytes from server-18-67-150-142.maa51.r.cloudfront.net (18.67.150.142): icmp_seq=3 ttl=242 time=19.1 ms
```

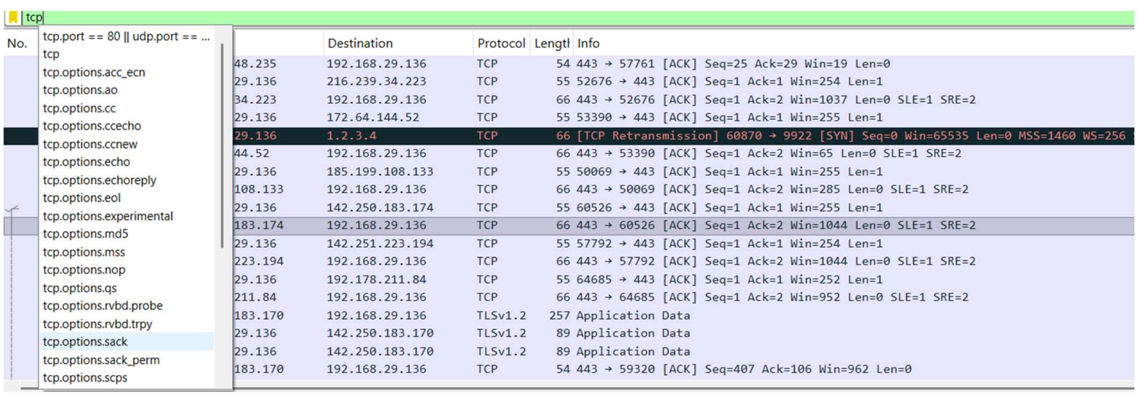
2.



```
udp
No.  udp      Source           Destination      Protocol Lengtl Info
     udpcp
     udpencap
     udplite
     ...875700 142.251.222.142  192.168.29.136   UDP      68 443 → 64921 Len=26
     ...174600 192.168.29.136   142.251.223.99   QUIC   1292 Initial, DCID=ed67f04f0fa953bd, PKN: 1, PING, CRYPTO, CRYPTO, PING, PIN
     ...564400 192.168.29.136   142.251.223.99   QUIC   1292 Initial, DCID=ed67f04f0fa953bd, PKN: 2, CRYPTO, CRYPTO, PING, CRYPTO
 138 12.468036700 192.168.29.136   142.251.223.99   QUIC    117 0-RTT, DCID=ed67f04f0fa953bd
 139 12.469448700 192.168.29.136   142.251.223.99   QUIC    956 0-RTT, DCID=ed67f04f0fa953bd
 140 12.514830300 142.251.223.99   192.168.29.136   QUIC     82 Initial, SCID=ed67f04f0fa953bd, PKN: 1, ACK
 141 12.516354200 142.251.223.99   192.168.29.136   QUIC   1292 Initial, SCID=ed67f04f0fa953bd, PKN: 2, ACK, PADDING
 142 12.519452700 142.251.223.99   192.168.29.136   QUIC   1292 Initial, SCID=ed67f04f0fa953bd, PKN: 3, CRYPTO, PADDING
 143 12.519452700 142.251.223.99   192.168.29.136   QUIC    326 Protected Payload (KP0)
 144 12.519452700 142.251.223.99   192.168.29.136   QUIC    986 Protected Payload (KP0)
 145 12.519452700 142.251.223.99   192.168.29.136   QUIC     67 Protected Payload (KP0)
 146 12.519592500 142.251.223.99   192.168.29.136   QUIC     64 Protected Payload (KP0)
 147 12.520433700 192.168.29.136   142.251.223.99   QUIC    120 Handshake, DCID=ed67f04f0fa953bd
 148 12.520644300 192.168.29.136   142.251.223.99   QUIC     73 Protected Payload (KP0), DCID=ed67f04f0fa953bd
 149 12.550176800 192.168.29.136   142.251.223.99   QUIC     74 Protected Payload (KP0), DCID=ed67f04f0fa953bd
 150 12.567246800 142.251.223.99   192.168.29.136   QUIC    162 Protected Payload (KP0)
 151 12.594364800 142.251.223.99   192.168.29.136   QUIC    559 Protected Payload (KP0)
 152 12.595157000 192.168.29.136   142.251.223.99   QUIC     77 Protected Payload (KP0), DCID=ed67f04f0fa953bd
```

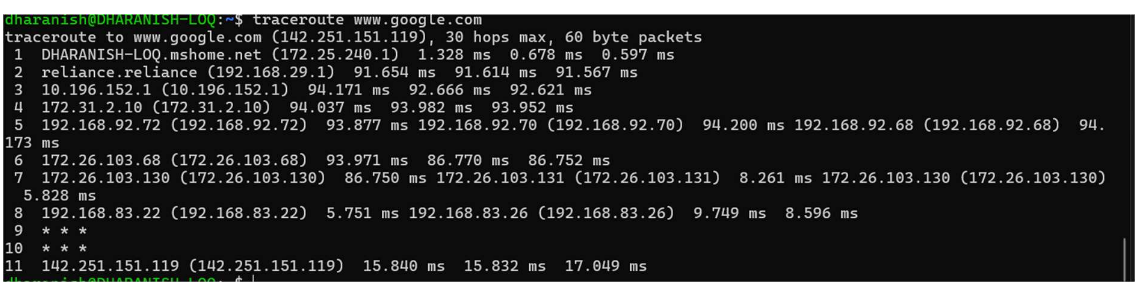


```
dns
No.  dns        Source           Destination      Protocol Lengtl Info
     dnsserver
     ...7122900 192.168.29.136   192.168.29.1     DNS       79 Standard query 0xa829 A clients4.google.com
 227 21.528684500 192.168.29.1     192.168.29.136   DNS      119 Standard query response 0xa829 A clients4.google.com CNAME clients.l.google.com A 172.217.26.46
 379 35.572178400 192.168.29.136   192.168.29.1     DNS       74 Standard query 0x25ce A www.google.com
 380 35.577489400 192.168.29.1     192.168.29.136   DNS      202 Standard query response 0x25ce A www.google.com A 142.251.156.119 A 142.251.155.119 A 142.251.154.119 A 142.:
 399 37.271958400 192.168.29.136   192.168.29.1     DNS       71 Standard query 0x4914 A chatgpt.com
 403 37.283986400 192.168.29.1     192.168.29.136   DNS      103 Standard query response 0x4914 A chatgpt.com A 172.64.155.209 A 104.18.32.47
 433 37.316539100 192.168.29.136   192.168.29.1     DNS       76 Standard query 0x37f8 A beacons.gvt2.com
 434 37.317948100 192.168.29.136   192.168.29.1     DNS       80 Standard query 0x816d A beacons.gcp.gvt2.com
 447 37.320718300 192.168.29.1     192.168.29.136   DNS      138 Standard query response 0x816d A beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com A 142.250.122.94
 458 37.336394700 192.168.29.1     192.168.29.136   DNS       92 Standard query response 0x37f8 A beacons.gvt2.com A 142.250.122.94
 545 43.463981500 192.168.29.136   192.168.29.1     DNS      116 Standard query 0xfbb1 A prod-dynamite-prod-07-us-signaler-pa.clients6.google.com
 550 43.503073200 192.168.29.136   192.168.29.1     DNS      116 Standard query 0xfbb1 A prod-dynamite-prod-07-us-signaler-pa.clients6.google.com
 551 43.532835400 192.168.29.1     192.168.29.136   DNS      132 Standard query response 0xfbb1 A prod-dynamite-prod-07-us-signaler-pa.clients6.google.com A 142.250.193.74
 559 43.532835400 192.168.29.1     192.168.29.136   DNS      132 Standard query response 0xfbb1 A prod-dynamite-prod-07-us-signaler-pa.clients6.google.com A 142.250.193.74
 729 54.493764100 192.168.29.136   192.168.29.1     DNS       94 Standard query 0x45bb A word-telemetry.officeapps.live.com
 731 54.500601200 192.168.29.1     192.168.29.136   DNS      201 Standard query response 0x45bb A word-telemetry.officeapps.live.com CNAME word-telemetry.wac.trafficmanager.r
 733 54.542033600 192.168.29.136   192.168.29.1     DNS       86 Standard query 0x123a A waa-pa.clients6.google.com
 737 54.562939500 192.168.29.136   192.168.29.1     DNS       84 Standard query 0xc708 A common.online.office.com
```



```
tcp
No.  tcp.port == 80 || udp.port == ...   Destination      Protocol Lengtl Info
     tcp                        48.235   192.168.29.136   TCP      54 443 → 57761 [ACK] Seq=25 Ack=29 Win=19 Len=0
     tcp.options.acc_ecn        29.136   216.239.34.223   TCP      55 52676 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
     tcp.options.ao             34.223   192.168.29.136   TCP      66 443 → 52676 [ACK] Seq=1 Ack=2 Win=1037 Len=0 SLE=1 SRE=2
     tcp.options.cc             29.136   172.64.144.52    TCP      55 53390 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
     tcp.options.ccecho         29.136   1.2.3.4          TCP      66 [TCP Retransmission] 60870 → 9922 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256
     tcp.options.ccnew          44.52    192.168.29.136   TCP      66 443 → 53390 [ACK] Seq=1 Ack=2 Win=65 Len=0 SLE=1 SRE=2
     tcp.options.echo           29.136   185.199.108.133  TCP      55 50069 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
     tcp.options.echoreply      108.133  192.168.29.136   TCP      66 443 → 50069 [ACK] Seq=1 Ack=2 Win=285 Len=0 SLE=1 SRE=2
     tcp.options.eol            29.136   142.250.183.174  TCP      55 60526 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
     tcp.options.experimental   183.174  192.168.29.136   TCP      66 443 → 60526 [ACK] Seq=1 Ack=2 Win=1044 Len=0 SLE=1 SRE=2
     tcp.options.md5            29.136   142.251.223.194  TCP      55 57792 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
     tcp.options.mss            223.194  192.168.29.136   TCP      66 443 → 57792 [ACK] Seq=1 Ack=2 Win=1044 Len=0 SLE=1 SRE=2
     tcp.options.nop            29.136   192.178.211.84   TCP      55 64685 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
     tcp.options.qs             211.84   192.168.29.136   TCP      66 443 → 64685 [ACK] Seq=1 Ack=2 Win=952 Len=0 SLE=1 SRE=2
     tcp.options.rvbd.probe     183.170  192.168.29.136   TLSv1.2 257 Application Data
     tcp.options.rvbd.trpy      29.136   142.250.183.170  TLSv1.2  89 Application Data
     tcp.options.sack           29.136   142.250.183.170  TLSv1.2  89 Application Data
     tcp.options.sack_perm      183.170  192.168.29.136   TCP      54 443 → 59320 [ACK] Seq=407 Ack=106 Win=962 Len=0
     tcp.options.scps
```

3.



```
dharanish@DHARANISH-LOQ:~$ traceroute www.google.com
traceroute to www.google.com (142.251.151.119), 30 hops max, 60 byte packets
 1  DHARANISH-LOQ.mshome.net (172.25.240.1)  1.328 ms  0.678 ms  0.597 ms
 2  reliance.reliance (192.168.29.1)  91.654 ms  91.614 ms  91.567 ms
 3  10.196.152.1 (10.196.152.1)  94.171 ms  92.666 ms  92.621 ms
 4  172.31.2.10 (172.31.2.10)  94.037 ms  93.982 ms  93.952 ms
 5  192.168.92.72 (192.168.92.72)  93.877 ms 192.168.92.70 (192.168.92.70)  94.200 ms 192.168.92.68 (192.168.92.68)  94.
173 ms
 6  172.26.103.68 (172.26.103.68)  93.971 ms  86.770 ms  86.752 ms
 7  172.26.103.130 (172.26.103.130)  86.750 ms 172.26.103.131 (172.26.103.131)  8.261 ms 172.26.103.130 (172.26.103.130)
 5.828 ms
 8  192.168.83.22 (192.168.83.22)  5.751 ms 192.168.83.26 (192.168.83.26)  9.749 ms  8.596 ms
 9  * * *
10  * * *
11  142.251.151.119 (142.251.151.119)  15.840 ms  15.832 ms  17.049 ms
```

5.

```
Switch>
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name CSE
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name II
Switch(config-vlan)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface fa0/2
Switch(config-if)#switchport access vlan 10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#end
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- ----------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, F
                                                Fa0/7, Fa0/8, Fa0/9, F
                                                Fa0/11, Fa0/12, Fa0/13
                                                Fa0/15, Fa0/16, Fa0/17
                                                Fa0/19, Fa0/20, Fa0/21
                                                Fa0/23, Fa0/24, Gig0/1
10   CSE                              active    Fa0/1
20   II                               active    Fa0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

```
Switch>
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------
1    default                          active    Fa0/1,
                                                Fa0/5,
                                                Fa0/9,
                                                Fa0/13,
                                                Fa0/17,
                                                Fa0/21,
                                                Gig0/1,
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#configure terminal
Enter configuration commands, one per line.  End with C
Switch(config)#vlan 10
Switch(config-vlan)#name CSE
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name II
Switch(config-vlan)#exit
Switch(config)#interface fa0/1
```

```
Switch>
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name CSE
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name II
Switch(config-vlan)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface fa0/2
Switch(config-if)#switchport access vlan 10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
```

```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)# vlan 10
Switch(config-vlan)#name BIO
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name EEE
Switch(config-vlan)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode acess
                                     ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport mode access
Switch(config-if)#switchport acess vlan 10
                                ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
```

```
C:\>ping 172.16.16.240

Pinging 172.16.16.240 with 32 bytes of data:


Ping statistics for 172.16.16.240:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),

Control-C
^C
C:\>ping 172.16.16.220

Pinging 172.16.16.220 with 32 bytes of data:


Ping statistics for 172.16.16.220:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),

Control-C
^C
C:\>ping 172.16.16.240
```

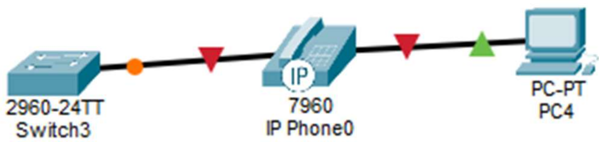
6.

```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk native vlan 10
                                        ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport trunk native vlan 10
Switch(config-if)#show interfaces trunk
                       ^
% Invalid input detected at '^' marker.

Switch(config-if)#exit
Switch(config)#show interfaces trunk
                    ^
% Invalid input detected at '^' marker.

Switch(config)#
```

7.

```
Switch(config-vlan)#name mgmnt
Switch(config-vlan)#interface vlan 99
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan99, changed state to up
ip address 192.168.99.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#interface fa0/1
Switch(config-if)#switchport mode acess
                                   ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 99
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up

Switch(config-if)#ip default-gateway 192.168.99.254
Switch(config)#hostname Switch
Switch(config)#ip domain-name lab.com
Switch(config)#crypto key generate rsa
% Please define a hostname other than Switch.
Switch(config)#username admin password 1234
Switch(config)#
Switch(config)#line vty 0 4
Switch(config-line)#login local
Switch(config-line)#transport input ssh
Switch(config-line)#hostname switches
switches(config)#ip domain-name lab.com
switches(config)#crypto key generate rsa
The name for the keys will be: switches.lab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

switches(config)#username admin password 1234
*Mar 1 0:3:30.595: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:3:30.595: %SSH-5-ENABLED: SSH 1.5 has been enabled
switches(config)#line vty 0 4
switches(config-line)#login local
switches(config-line)#transport input ssh
switches(config-line)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (1), with Switch
FastEthernet0/1 (10).
```

8.




```
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#switchport voice vlan 20
Switch(config-if)#end
Switch#show vlan bried
                     ^
% Invalid input detected at '^' marker.

Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   data                             active    Fa0/1
20   ip                               active    Fa0/1
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#show interfaces fa0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: down
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 10 (data)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: 20
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
 --More-- |
```

enable

configure terminal

access-list 1 deny host 192.168.20.11

access-list 1 permit any

access-list 101 deny tcp any any eq 80

access-list 101 permit ip any any

access-list 101 deny tcp 192.168.10.0 0.0.0.255 host 192.168.20.10 eq 80

access-list 101 permit ip any any

FTP control port is 21.

access-list 102 deny tcp 192.168.10.0 0.0.0.255 host 192.168.20.10 eq 21

access-list 102 permit ip any any

15.

```
dharanish@DHARANISH-LOQ:~$ iperf3 -c  172.25.253.115
Connecting to host 172.25.253.115, port 5201
[  5] local 172.25.253.115 port 38304 connected to 172.25.253.115 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec  6.68 GBytes  57.4 Gbits/sec    0   3.18 MBytes
[  5]   1.00-2.00   sec  6.71 GBytes  57.6 Gbits/sec    0   3.37 MBytes
[  5]   2.00-3.00   sec  6.65 GBytes  57.1 Gbits/sec    1   3.93 MBytes
[  5]   3.00-4.00   sec  6.71 GBytes  57.6 Gbits/sec    0   3.93 MBytes
[  5]   4.00-5.00   sec  6.73 GBytes  57.8 Gbits/sec    0   3.93 MBytes
[  5]   5.00-6.00   sec  6.72 GBytes  57.7 Gbits/sec    0   3.93 MBytes
[  5]   6.00-7.00   sec  6.62 GBytes  56.8 Gbits/sec    0   3.93 MBytes
[  5]   7.00-8.00   sec  6.80 GBytes  58.4 Gbits/sec    0   3.93 MBytes
[  5]   8.00-9.00   sec  6.71 GBytes  57.7 Gbits/sec    0   3.93 MBytes
[  5]   9.00-10.00  sec  6.66 GBytes  57.2 Gbits/sec    0   3.93 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-10.00  sec  67.0 GBytes  57.5 Gbits/sec    1             sender
[  5]   0.00-10.04  sec  67.0 GBytes  57.3 Gbits/sec                  receiver

iperf Done.
dharanish@DHARANISH-LOQ:~$ iperf3 -c  172.25.253.115 -u
Connecting to host 172.25.253.115, port 5201
[  5] local 172.25.253.115 port 37220 connected to 172.25.253.115 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   128 KBytes  1.05 Mbits/sec  4
[  5]   1.00-2.00   sec   128 KBytes  1.05 Mbits/sec  4
[  5]   2.00-3.00   sec   128 KBytes  1.05 Mbits/sec  4
[  5]   3.00-4.00   sec   128 KBytes  1.05 Mbits/sec  4
[  5]   4.00-5.00   sec   128 KBytes  1.05 Mbits/sec  4
[  5]   5.00-6.00   sec   128 KBytes  1.05 Mbits/sec  4
[  5]   6.00-7.00   sec   128 KBytes  1.05 Mbits/sec  4
[  5]   7.00-8.00   sec   128 KBytes  1.05 Mbits/sec  4
[  5]   8.00-9.00   sec   128 KBytes  1.05 Mbits/sec  4
[  5]   9.00-10.00  sec   128 KBytes  1.05 Mbits/sec  4
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  1.25 MBytes  1.05 Mbits/sec  0.000 ms  0/40 (0%)  sender
[  5]   0.00-10.04  sec  1.25 MBytes  1.04 Mbits/sec  0.082 ms  0/40 (0%)  receiver

iperf Done.
dharanish@DHARANISH-LOQ:~$ iperf3 -c  172.25.253.115 -i 10
Connecting to host 172.25.253.115, port 5201
[  5] local 172.25.253.115 port 56434 connected to 172.25.253.115 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-10.00  sec  65.6 GBytes  56.3 Gbits/sec    0   4.00 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-10.00  sec  65.6 GBytes  56.3 Gbits/sec    0             sender
[  5]   0.00-10.04  sec  65.6 GBytes  56.1 Gbits/sec                  receiver

iperf Done.
dharanish@DHARANISH-LOQ:~$ 172.25.253.115
```